

ADVERTISING MONETISATION MANAGER TEST

General Instructions

1. Estimated time: up to 4 hours of written work, plus about 30 minutes playing the game. If you run short, prioritise Sections 2 and 3.
2. Allowed tools: Excel, Google Sheets, Python or R. AI assistants (ChatGPT, Claude, Copilot and similar) are allowed too; note where you used one and what you checked yourself. The final presentation (20 minutes, followed by questions) will test your reasoning, so be ready to defend every number and every judgement.
3. Submit a Word document, PDF or Notion link to the recruiting team within 24 hours of receiving this document. The window is deliberately wider than the work. If it is impossible for you, tell us early rather than sending something rushed.
4. All data is illustrative and was created for this exercise. It does not represent real business performance.
5. The open questions have no single correct answer. We are as interested in **how you reason** as in the numbers, so show your working and state your assumptions.
6. Play **Enigmatic: Crime Puzzle Cases** before you answer: [iOS](#) / [Android](#).

How We Will Evaluate

- **Numerical accuracy:** correct, reproducible calculations, assumptions stated.
- **Commercial judgement:** balancing short-term revenue against retention and LTV.
- **Experiment design:** confounds, sample sizes, what the test can support.
- **Mediation knowledge:** waterfalls, bidding, fill and show rates, in practice.
- **Communication:** clear, concise, and structured for a non-specialist audience.

Section 1: Concepts & Judgement

Short answers, a paragraph or a few bullets each, with Enigmatic in mind. We are assessing exposure and instincts, not textbook recall: a concrete example from your own experience beats a definition. **Four** further topics are for the final interview rather than this document, so it is worth having a view on them: how an ad request becomes a paid impression in a mediated setup; waterfall versus in-app bidding, and why most games run a hybrid; why opted-out iOS traffic monetises at a lower eCPM, and what a monetisation manager can do about it on the supply side; and what UA needs on the data side to bid on D7 ad ROAS.

1.1 Where ad ARPDAU comes from. Ad ARPDAU can be written as: $(\text{share of DAU that sees at least one ad}) \times (\text{impressions per ad-viewer}) \times (\text{eCPM} \div 1,000)$. For each of the three factors, give one lever you control and the main risk of pulling it — **one line each**. Then say which factor you would look at first for a casual puzzle game, and why.

1.2 Floor vs reality. One of our networks reports an **actual eCPM roughly 40% below its floor price**. Give three possible explanations and describe how you would tell them apart. Separately, the same network's own dashboard shows **8% more revenue** than our mediation platform reports for it: name the two most likely causes and say which you would check first.

Section 2: Placement Performance

Your goal is to work out how each ad placement in Enigmatic is actually performing, and where the next impressions should come from.

The data covers the 7 days from **24.08.2026 to 30.08.2026** (inclusive). Average DAU over the period was **96,000**.

Placement	Format	Where it shows
Case complete	Interstitial	After a completed case, frequency-capped
Reveal a clue	Rewarded	Clue button inside a case
App open	Interstitial	On app launch

Placement data, 24.08.2026 to 30.08.2026 (7-day totals; viewers are a daily average)

Placement	Ad requests	Filled	Impressions	Revenue (\$)	Daily viewers
Case complete	2,000,000	1,920,000	1,680,000	16,800.00	60,000
Reveal a clue	1,280,000	1,152,000	604,800	15,120.00	21,600
App open	720,000	672,000	537,600	3,225.60	76,800

Definitions

- **Ad request:** the app asked mediation for an ad. **Filled:** it returned one. **Impression:** the ad rendered on screen and was paid for.
- **Fill rate** = filled ÷ ad requests. **Show rate** = impressions ÷ filled. Use these two definitions throughout Section 2, so your numbers are comparable with ours.
- **Daily viewers:** distinct users per day who saw at least one impression from the placement (daily average). **Imp/DAU:** impressions per daily active user per day.

2.1 Placement scorecard. For each placement, calculate the **show rate**, **imp/DAU**, **share of DAU reached per day** and **eCPM**. Then give **ad ARPDAU** and **eCPM** for the game as a whole. Show your formulas. You do not need to calculate a fill rate for every placement — just note anything about fill that stands out, and note anything that cannot be calculated at game level from this table.

2.2 Wasted loads. One placement turns far fewer filled ads into impressions than the others. Calculate the weekly revenue upside if its show rate rose to **90%** (a deliberately optimistic ceiling), with every recovered fill paid at today's eCPM. Then give two different causes that would produce its numbers, say what extra data would tell them apart, and explain why the real upside would be lower than your figure.

2.3 Where the next impressions come from. Product has proposed two changes and can only ship one this quarter:

Proposal	Expected effect
Case-abandon interstitial: a new interstitial when a player abandons a case	Adds 1.0 imp/DAU at the Case complete eCPM
Clue offer on the case screen: the Reveal a clue offer surfaced on the case screen itself	Raises that placement's share of DAU reached from 22.5% to 40% , with impressions rising in proportion to reach

- Calculate the ad ARPDAU impact of each at today's eCPMs, say which one you would ship, and use the three factors from question 1.1 to name the risk you are taking.
- Having played Enigmic: would you back either of these, or propose something else? Name one placement you would add, remove or change, and one thing about the game itself (session length, difficulty curve, progression) that limits how much ad inventory it can carry. Two or three sentences on each is enough.

Section 3: A/B Test Analysis

A **7-day** A/B test on Enigmic compares two ad monetisation setups, split **50/50 at assignment**. Both variants ran the same networks, the same mediation setup and the same price floors: the only differences are the two in the table below. Installs: **73,500** in A and **73,080** in B, spread evenly over the week; retention refers to them. The data is independent of Section 2. Changes are given so you can spend your time on what they mean rather than on the arithmetic.

Setup	Variant A (Control)	Variant B (Test)
Interstitials	One after every 3 completed cases	One after every 2 completed cases
Rewarded	Existing placements only	A new rewarded placement added: "Reveal a second clue", offered on the case-failed screen

Dataset: 7-day totals and averages

Metric	Variant A (Control)	Variant B (Test)	Change
Average DAU	84,000	83,200	-1.0%
Total ad impressions	2,646,000	3,640,000	+37.6%
Imp/DAU	4.50	6.25	+38.9%
Total ad revenue (\$)	35,721	44,772	+25.3%
eCPM (\$)	13.50	12.30	-8.9%
Ad ARPDAU (\$)	0.0608	0.0769	+26.5%
Total IAP revenue (\$)	8,400	6,700	-20.2%
Average session length (min)	12.4	10.9	-12.1%
Sessions per DAU per day	3.2	2.9	-9.4%
Cases completed per DAU per day	9.1	7.4	-18.7%
D1 retention	44%	41%	-3 pp
D7 retention	21%	17%	-4 pp
% of DAU watching at least 1 rewarded ad	23%	32%	+9 pp

Metric	Variant A (Control)	Variant B (Test)	Change
Support tickets mentioning ads	57	211	+270%
New 1-star store reviews	36	149	+314%

3.1 Revenue, exposure and price

- The table gives the change for every metric. The split was **50/50 at assignment**, so what does the DAU gap tell you? And nothing about how ads are priced or sold changed between the variants, so why did **eCPM** fall?
- Ad ARPDAU is given in the table. Calculate **total ARPDAU** (ads + IAP) for both variants. Why does total revenue move less than ad revenue, and which of the two should we steer on?

3.2 User experience and retention

- Do the engagement and quality changes point to a worse player experience? Which metrics in the table are leading indicators rather than lagging ones?
- Are the retention differences large enough to be confident they are real? Use the install figures to work out how many users each retention figure rests on, and show your working.

3.3 Long-term value. As a first approximation, take LTV per install as proportional to **total ARPDAU × D7 retention**. Calculate the percentage change in LTV from Variant A to Variant B, and therefore in the **maximum CPI** the UA team can afford to pay for an install. Say briefly why this is only an approximation.

3.4 Recommendation and next test

- Write the recommendation you would send to the product lead: ship **Variant B**, stay on **Variant A**, or iterate. Lead with the three numbers from the table that decide it, and say what about this test's design limits what you can conclude from it.
- Whatever you recommend, design the next test in a few lines: what changes, the primary metric, the guardrail metrics that would stop it, and how long it runs.

Good luck! 😊

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Content

Section 1: Concepts & Judgement	1
Where ad ARPDau comes from	1
Share of DAU seeing an ad	1
Impressions per ad-viewer	1
Revenue per impression	1
Floor vs reality	1
Actual eCPM is 40% below the floor	1
Network dashboard shows 8% more revenue than mediation	2
Section 2: Placement Performance	2
Placement Scorecard	2
<i>Calculations*</i>	2
Wasted Loads – Reveal a Clue	3
<i>Calculations*</i>	3
Where the next impressions come from	4
<i>Calculations*</i>	4
Section 3: A/B Test Analysis	5
Revenue, exposure and price	5
<i>Calculations*</i>	5
User Experience and Retention	5
<i>Calculations*</i>	6
Long-term value	6
<i>Calculations*</i>	6
Recommendation and next step	7

Section 1: Concepts & Judgement

Where ad ARPDAU comes from

Share of DAU seeing an ad

Lever: move ads early in the session (or even app launch), at a natural break such as after a completed case.

Risk: retention damage if not enough value is perceived before running into an ad.

Impressions per ad-viewer

Lever: increase ad frequency (either reducing interstitial cooldown or create rewarded ads inventory – around hints for ex.)

Risk: ad fatigue, shorter sessions and ultimately fewer lifetime impressions

Revenue per impression

Lever: improve auction pressure by adding or optimizing demand partners and testing floors for high volume segments.

Risk: sacrificing total revenue if higher yield comes at the expense of fill rate. as well as operational complexity by managing too many partners.

For Enigmatic I would first decompose all three factors by platform, geo and format, but my initial hypothesis would be to go for eCPM optimization. A casual puzzle game has natural breaks and if engagement is high enough it should generate enough impressions volume. If reach and impressions are healthy but ARPDAU is low, increasing ad frequency could hurt retention without addressing the underlying yield problem.

Floor vs reality

Actual eCPM is 40% below the floor

- First, the floor may be soft or dynamically relaxed rather than a guaranteed minimum. I would confirm the floor type and inspect auction data – below floor bids being accepted on requests covered by this rule would support this explanation. Second, the reported actual eCPM may include traffic outside the floor's scope. I would filter both reports by the same ad unit, placement, platform, geo and mediation. If the filtered eCPM meets the floor aggregation was the issue. And then third, reporting quality – the two values may use different reporting definitions such as currency, time zone or gross vs net revenue. If normalizing these values and recalculating eCPM from revenue and impressions clears the gap it can be cleared as a reporting issue.

Network dashboard shows 8% more revenue than mediation

- As mentioned previously, one of the main causes of this discrepancy can be **reporting differences** – finalization differences across time zones, reporting delays or revenue being estimated at different times - or **traffic integration/scope** – non mapped traffic being included in network's dash, ad units not included or wrongly included (legacy or test ad units), missed mediation traffic going straight to network's data or API missing revenue. I would first align the date range, time zone, currency, ad units and revenue status in both systems to clear common mismatches. Then focus on impressions, if impressions count match but revenue does not, the issue is likely currency, revenue finalization or gross vs net treatment. If impressions also differ, I would investigate ad unit mapping, direct network traffic and overall mediation reporting.

Section 2: Placement Performance

Placement Scorecard

*Calculations**

Fill rate = Filled / Ad requests Show rate = Impressions / Filled
Impressions/day = Impressions / Days Imp/DAU = (Impressions/day) / DAU
Share of DAU reached = Daily viewers / DAU eCPM = (Revenue / Impressions) x 1,000

Case complete:

Fill rate = 1,920,000 / 2,000,000 = 96.0%
Show rate = 1,680,000 / 1,920,000 = 87.5%
Impressions/day = 1,680,000 / 7 = 240,000
Imp/DAU = 240,000 / 96,000 = 2.50
Share of DAU reached = 60,000 / 96,000 = 62.5%
eCPM = (16,800.00 / 1,680,000) x 1,000 = \$10.00

Reveal a clue:

Fill rate = 1,152,000 / 1,280,000 = 90.0%
Show rate = 604,800 / 1,152,000 = 52.5%
Impressions/day = 604,800 / 7 = 86,400
Imp/DAU = 86,400 / 96,000 = 0.90
Share of DAU reached = 21,600 / 96,000 = 22.5%
eCPM = (15,120.00 / 604,800) x 1,000 = \$25.00

App open:

Fill rate = 672,000 / 720,000 = 93.3%
Show rate = 537,600 / 672,000 = 80.0%
Impressions/day = 537,600 / 7 = 76,800
Imp/DAU = 76,800 / 96,000 = 0.80

Share of DAU reached = $76,800 / 96,000 = 80.0\%$
eCPM = $(3,225.60 / 537,600) \times 1,000 = \6.00

Game-level (whole game):

Total revenue = $16,800.00 + 15,120.00 + 3,225.60 = \$35,145.60$

Total impressions = $1,680,000 + 604,800 + 537,600 = 2,822,400$

Game eCPM = $(35,145.60 / 2,822,400) \times 1,000 = \12.45

Game Ad ARPDAU = Total revenue / (DAU x Days) = $35,145.60 / (96,000 \times 7) = \0.0523

Game Imp/DAU = Total impressions / (DAU x Days) = $2,822,400 / 672,000 = 4.20$

Summed fill rate = Total filled / Total requests = $3,744,000 / 4,000,000 = 93.6\%$

- a) Although Fill rate is showing balanced across placements, meaning the ads requested are being delivered for the most part, "Reveal a clue" placement Show rate is significantly lower compared with remainder placements which means ads are ready but not being delivered - ex: players backing out before the video play. This, pairing with an also lower share of DAU reached per day, points to a placement converting very few of the requested adds into actual paid impressions. It is also the highest eCPM across placements, which makes it the most expensive gap to address
- b) True reach cannot be calculated based on provided data. The daily viewers numbers add up to 158 400 but there is only 96 000 DAU, so there are more viewers being counted across more than one placement. Without user level - instead of pre-aggregated - data to apply a distinct count on a game-wide reach figure is not possible. All other metrics in the table are count of events, so it is ok.

Wasted Loads – Reveal a Clue

Calculations*

Current show rate = Impressions / Filled = $604,800 / 1,152,000 = 52.5\%$

New impressions @ 90% show rate = Filled x 90% = $1,152,000 \times 0.90 = 1,036,800$

Recovered fills = New impressions - Current impressions = $1,036,800 - 604,800 = 432,000$

Weekly revenue upside = (Recovered fills / 1,000) x eCPM = $(432,000 / 1,000) \times \$25.00 = \$10,800.00$

- a) Two causes that would produce this uplift could be connected to friction or technical change. Friction change if there is a change in the flow removing a step before watching the ad - at the moment a window is presented when running out of hints that includes a prompt to watch an ad to get more hints - removing this step between using a hint and watching the ad can increase show rate. It can also be ad load latency that players won't wait through, although personally I did not experience any significant load latency myself. Technical cause could point to the ad being loaded beforehand but not being shown when called upon, either because the ad expired or was not ready at the time it was called upon.

- b) Extra data required would be precisely *load()* and *show()* telemetry events. Did the ad load? Was it shown? If not, what was the logged error? Also, if we are able to see if the player left the case screen before the ad started or during the ad confirmation step we can address it as a flow fix, instead of technical.
- c) The real upside would be lower than \$10,800 because the actual eCPM applied for would be lower to some of the newly added impressions due to bidding setup or top networks not being able to handle the extra volume at the defined eCPM of \$25,00. Apart from that, there is also a share of players that won't get the ad shown effectively either because the ad is too long and players quit halfway through connection issues or old devices that fail at a baseline rate.

Where the next impressions come from

Calculations*

Proposal A - Case-abandon interstitial:

Ad ARPDAU impact = Added Imp/DAU x (eCPM / 1,000) = 1.00 x (\$10.00 / 1,000) = \$0.0100 /DAU/day

Weekly revenue impact = Ad ARPDAU impact x DAU x Days = \$0.0100 x 96,000 x 7 = \$6,720.00

Proposal B - Clue offer on case screen:

Reach ratio = New share / Current share = 40.0% / 22.5% = 1.78

New Imp/DAU = Current Imp/DAU x Reach ratio = 0.90 x 1.78 = 1.60

Delta Imp/DAU = New Imp/DAU - Current Imp/DAU = 1.60 - 0.90 = 0.70

Ad ARPDAU impact = Delta Imp/DAU x (eCPM / 1,000) = 0.70 x (\$25.00 / 1,000) = \$0.0175 /DAU/day

Weekly revenue impact = Ad ARPDAU impact x DAU x Days = \$0.0175 x 96,000 x 7 = \$11,760.00

Proposal B (\$11,760) > Proposal A (\$6,720) -> higher weekly revenue impact

- a) I would ship proposal B - mainly because option A, besides showing lower weekly impact, adds another interstitial moment to the same gameplay moment (case complete or case abandon), aggravated by being a bad moment - case abandon - so it compromises retention even further. Proposal B increases reach and being an opt-in ad format reduces the ad fatigue feeling - although the risk here is exposing more, less segmented players to ad viewing options and can widen negative sentiment instead of concentrating it.
- b) Having played Enigmatic I would back Proposal B because of its in-context nature. The need is being flagged by the player and it's being met with the ad inside the context of the case. If the player is not too ad-verse, it's a fair tradeoff. One placement I would change is Case Complete Interstitial. Particularly during D1 by keeping case complete interstitials on a 3 to 5 case interval cooldown period (pending testing). Would also avoid showing a forced interstitial after a rewarded ad. If the player knows a forced ad is coming at case complete, he/she is less likely to willingly add an ad on top of that. If the player notices that when watching a rewarded ad, a forced interstitial is not shown, the rewarded ad feels more welcoming.

One thing about the game I would address is being clearer around the value offer of progressing through the cases. Apart from per case completion and puzzle completion satisfaction, it doesn't seem to have a broader end goal. At the moment it seems the only way to climb through the detective ranks is by paying. But having time spent completing the case and (maybe) moves taken during case completion, a progression-based reward system could be built, making it possible for the player to grind its way towards higher ranks without having to pay. New inventory can be applied after celebration moments like reaching milestones.

Section 3: A/B Test Analysis

Revenue, exposure and price

- a) DAU gap tells me that although samples started roughly the same with around 73k installs, fewer players from the Test group showed up during the week. Which is backed up by D1 and D7 retention values also being lower for the Test group. eCPM changed due to the nature of the ads being served. Test group setup includes more frequent automatic interstitials, increasing impressions volume and per DAU. eCPM is shown as a blended average from both ad formats performance which is highly affected by which ad format got the most volume, even if prices stay the same. In this case, interstitial volume comes from every player whereas rewarded ad volume only comes from 32% of players that opted in. Also raising +38% total impressions while networks and floors stay fixed means increasing supply from the same demand pool and therefore accepting lower eCPM's biddings.

*Calculations**

DAU-days = DAU x Test length

Variant A: $84,000 \times 7 = 588,000$

Variant B: $83,200 \times 7 = 582,400$

IAP ARPDAU = Total IAP revenue / DAU-days

A: $8,400.00 / 588,000 = \$0.0143$

B: $6,700.00 / 582,400 = \$0.0115$

Total ARPDAU = Ad ARPDAU + IAP ARPDAU

A: $0.0608 + 0.0143 = \$0.0751$

B: $0.0769 + 0.0115 = \$0.0884$

% change = $(\text{Total ARPDAU}_B - \text{Total ARPDAU}_A) / \text{Total ARPDAU}_A$

= $(0.0884 - 0.0751) / 0.0751 = +17.7\%$

- b) Total revenue is changing less than ad revenue due to total IAP revenue dropping 20%. This most certainly is due to ads pulling players away from purchasing IAPs. We should steer towards ARPDAU. Focusing on ad ARPDAU may be moving revenue from one place to the other, it is not new revenue.

User Experience and Retention

- a) Most definitely points to a worst player experience. All engagement metrics are down – session length, sessions per DAU, cases completed per DAU, D1 and D7 retention – while support tickets and 1-star reviews skyrocketed. Leading indicators would be the immediate metrics such as session length, number of sessions, support tickets and star

**Calculations performed with*

Claude AI assistance and duly verified

Rodolfo Silva

5

reviews, whereas lagging metrics would be time bound, such as retention and revenue figures, ultimately driving LTV. Given that leading metrics are already bad, lagging metrics will likely get worse over time.

Calculations*

Installs/day = Total installs / 7

A: 73,500 / 7 = 10,500/day

B: 73,080 / 7 = 10,440/day

D1 cohort (>=1 day elapsed, i.e. days 1-6 of installs) = Installs/day x 6

A: 10,500 x 6 = 63,000

B: 10,440 x 6 = 62,640

D7 cohort (full 7 days elapsed, i.e. day-1 installs only) = Installs/day x 1

A: 10,500

B: 10,440

SE of difference = $\sqrt{p_A(1-p_A)/n_A + p_B(1-p_B)/n_B}$

D1: $p_A=44\%$, $n_A=63,000$; $p_B=41\%$, $n_B=62,640$

$SE(D1) = \sqrt{0.44 \times 0.56 / 63,000 + 0.41 \times 0.59 / 62,640} = 0.28pp$

Difference = 44% - 41% = 3.00pp

$z = \text{Difference} / SE = 3.00 / 0.28 = 10.76$ ($|z| > 1.96 \rightarrow$ significant at 95%)

D7: $p_A=21\%$, $n_A=10,500$; $p_B=17\%$, $n_B=10,440$

$SE(D7) = \sqrt{0.21 \times 0.79 / 10,500 + 0.17 \times 0.83 / 10,440} = 0.54pp$

Difference = 21% - 17% = 4.00pp

$z = \text{Difference} / SE = 4.00 / 0.54 = 7.39$ ($|z| > 1.96 \rightarrow$ significant at 95%)

- b) Yes, considering the percentage of both cohort D1 and D7 retention attributed to noise/randomness vs observable difference, retention figures are large enough to be confident on above 95% as real effects, even excluding D7 cohort impact.

Long-term value

Calculations*

LTV index = Total ARPDAU x D7 retention

A: $0.0751 \times 21\% = 0.01577$

B: $0.0884 \times 17\% = 0.01503$

% change in LTV = $(LTV_B - LTV_A) / LTV_A$

= $(0.01503 - 0.01577) / 0.01577 = -4.69\%$

Since max affordable CPI = $k \times LTV$ (breakeven logic, constant k):

% change in max CPI = % change in LTV = -4.69%

- a) Change in LTV per install from control to test group is set at (4.69%). Since CPI scales directly with LTV, the amount UA would be capped paying per install would also drop 4.69%. It is only an approximation because in fact compresses a lifetime revenue into one single retention point, it assumes verified ARPDAU stays the same across player's lifetime and it takes in population mismatch specific to this exercise where total ARPDAU is measured across whole DAU over the week (including returning or existing

players) and not just new installs, while D7 retention is specifically about the day 1 install cohort which refers to new installs only.

Recommendation and next step

- a) Recommendation would be not to ship Variant B as it is and not revert to Variant A. We should iterate on the Test group further given that APRDAU improved 17.7% which is good in isolation, but D7 retention dropped 4pp with real effect. Weighting revenue gain with retention loss, LTV per install drops 4.69%.
The test group inserted 2 changes at once with more frequent interstitials and a second rewarded ad placement, making it harder to isolate the effect of each. Although it can be assumed that increase in forced interstitials frequency risks retention loss, increase in support tickets and 1-star reviews.
Keeping rewarded ad second placement, being opt-in, can be a good next iteration when not placed alongside a change in interstitials frequency. Isolating rewarded ad second placement change may result in getting most of ad revenue upside without the retention drop.
- b) As stated before, what changes is de-bundle Variant B changes into one single change. Run the new rewarded ad placement on case failed, on its own, with interstitial frequency set at every 3 cases as it is. If sample size allows, a third variant can be set to test every 2 cases interstitial variant on its own as well. The idea is to not pull the two levers at the same time.
Primary metric would be LTV (with the same presented formula) with guardrail metrics being D1 and D7 retention, sessions length, sessions number and 1-star reviews mentioning ads, as well as supporting tickets.
I would increase the duration of this test to 14 days to allow more robust D7 figures and give more time to analyze IAP purchases trends that are not related to weekly peaks.